AI Agent for Medical Rehabilitation Coaching

Virtual Physical Therapy Assistant: Multimodal AI for At-Home Rehabilitation

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THE PROBLEM

~70-76%

Of patients are non-compliant with home rehab exercises

ONCE DISCHARGED

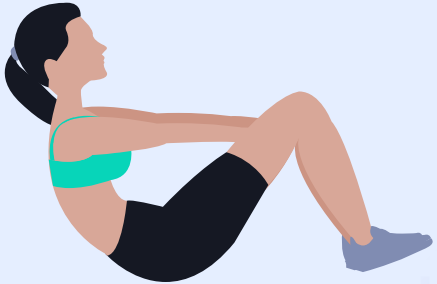
Patients lose the clinical supervisory feedback loop entirely

CONSEQUENCES

Compensatory movements, re-injury, prolonged recovery

PAIN-POINT

How to provide similar feedback experience at home vs in-clinic



OUR SOLUTION

VIRTUAL PHYSIOTHERAPY ASSISTANT

A multimodal AI agent that watches you exercise, identifies form errors, and coaches you — in real time, from your computer or phone.

COMPUTER VISION

Classifies exercises and mistakes using fine-tuned Mediapipe pose estimation

INTEGRATION LAYER

Cleans raw CV output, filters frames for coachable events and manages levels of coaching cues

RAG PIPELINE

Constructs clinically grounded feedback via RAG-enabled LLM and coaches your strengths and areas of improvement for each exercise

ARCHITECTURE DIAGRAM

CV model training
~80% accuracy
148 exercise classes

~290k exercise clips from QEVD dataset with annotations → MediaPipe (33 kpts · 15 fps) → Train PoseTCN: 8 multi-task heads, dilated causal TCN blocks → CV inference via webcam streaming at 15 fps → Streaming output: exercise, mistakes, quality score, metrics

Integration Layer
Cleans CV output
Manages coaching cues

Temporal aggregation: 10s window · ≥30% persistence · ≥3s duration · 10s cooldown → Coaching inputs: mistake types and severity, quality score, metrics

LangGraph

Risk routing & enrich context

Tier 1: low-risk or cached pattern → instant coaching, bypasses LLM · ~50ms

Tier 2: Claude Sonnet 4.5 → RAG grounded coaching cues · ~2–5s

Tier 3: AI agent → full reasoning for complex & high-risk mistakes· ~5–8s

Response Pipeline

Coaching agent: polish tone & style with patient personalization → Quality gate: length · relevance, Fallback feedback templates if low quality generation → Format + deliver: audio · UI display · session log and progress tracking

Front-End & Progress Tracker

Streamlit

Streamlit coaching front-end — live session feedback with CV skeleton overlay → Longitudinal Progress Tracker — trends, phase reports, LLM-generated rehab summaries

CV Pipeline: Pose Estimation & Feature Engineering

- Overlay Google MediaPipe pose estimation API over ~290k exercise videos
  - Capture stable pose landmarks once the system recognizes the user skeleton
  - Outputs 33 keypoints per frame and (x, y, z, visibility)
  - Needs to be a stable video, within certain jitter and glitching threshold to use
- Normalize, center to hips and make body and invariant system
- Videos convert to sequences of 198 feature vectors per frame
  - position (x, y) - 66 dims
  - velocity Δ(x, y) - 66 dims
  - depth z - 33 dims
  - confidence - 33 dims
- Store pose landmarks
  - Position, velocity, depth, and confidence score stored in .npz files
- Process to memmap (X.f16.mmap) for faster training and add back descriptive labels found in the original json description per video

CV Pipeline: PoseTCNTyped Model

- Input (T, 198) to Temporal Convolutional Neural Network and output multi head predictions
- 6 layers of 1d convolutions with kernel = 3 and exponential dilation
  - Captures short term dependency in shallow layers, and long term in deep layers
  - Classification and metrics form separate heads
  - Loss =
    - 1 * Exercise - categorical (softmax) - Cross Entropy loss
    - .8 * Mistakes - multi-label (sigmoid) - BCE loss
    - .3 * Speed - categorical or numeric - Cross Entropy loss
    - .3 * ROM / .2 *height / .2 *torso- ordinal classes - Cross Entropy loss
    - .2 * Direction - categorical - Cross Entropy loss
    - .2 * No issue - binary - BCE loss
- Optimizer - ADAMW, lr = 1e-4, weight decay = 1e-4, gradient clipping = 1.0
- Trained on random clips of varying length, with varying pose and camera angle per video

LLM Evaluation & Model Selection

- 3 models evaluated on 28 expert-authored physiotherapy questions
- Evaluation metrics: fact coverage, semantic coverage, completeness, safety pass rate, professional referral rate, medical accuracy, latency, cost

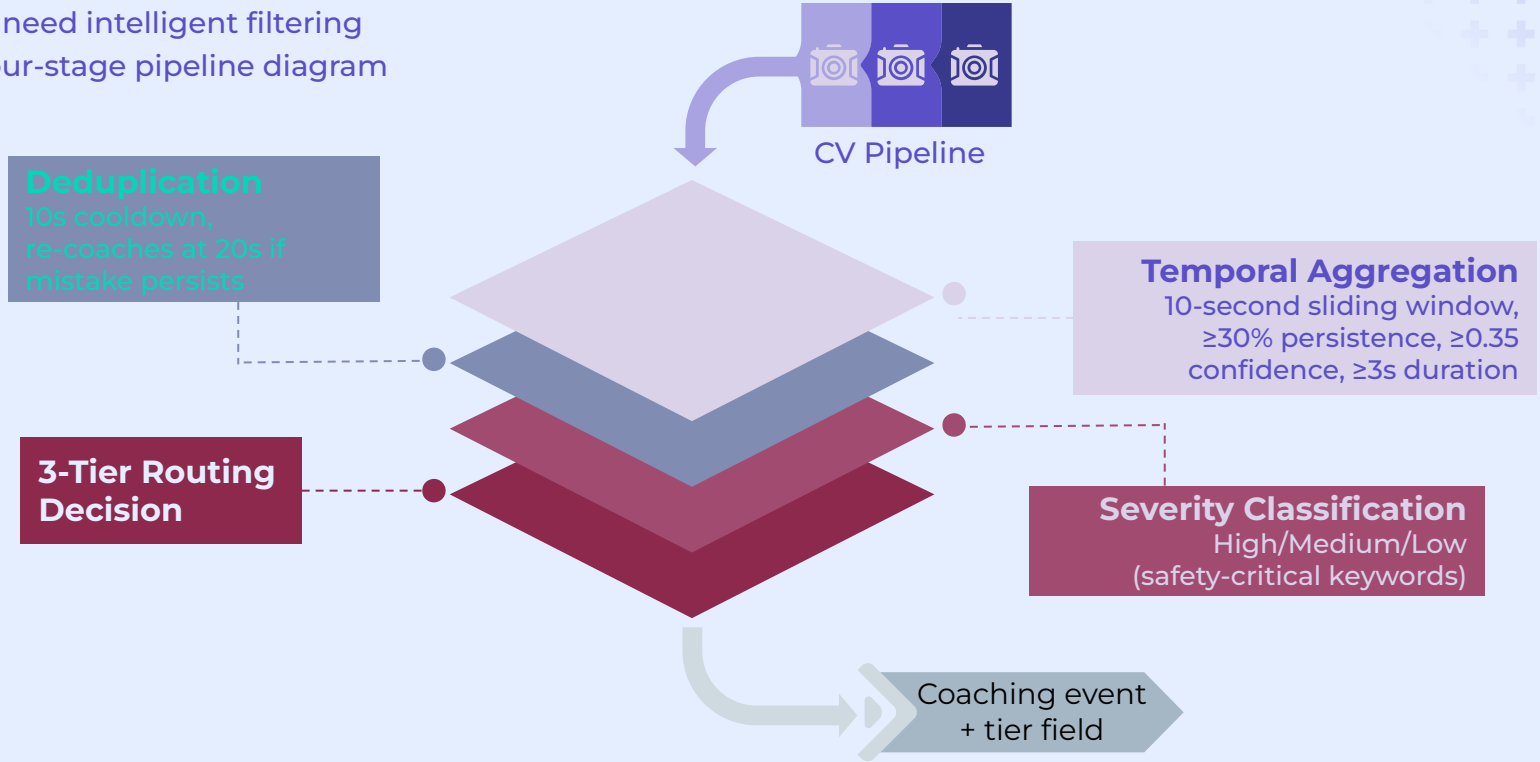
Model	Overall*	Fact Cov.	Sem Cov.	Completeness	Latency	Cost**
Claude Sonnet 4.5	**58.93**	**41.5%**	**69.7%**	**46.9%**	8.3s	$0.244
Qwen2.5:7b	54.18	31.4%	53.1%	35.7%	13.5s	$0
Gemma3:4b	54.04	29.8%	47.6%	34.1%	12.3s	$0

*Score out of 100
**In USD ($)

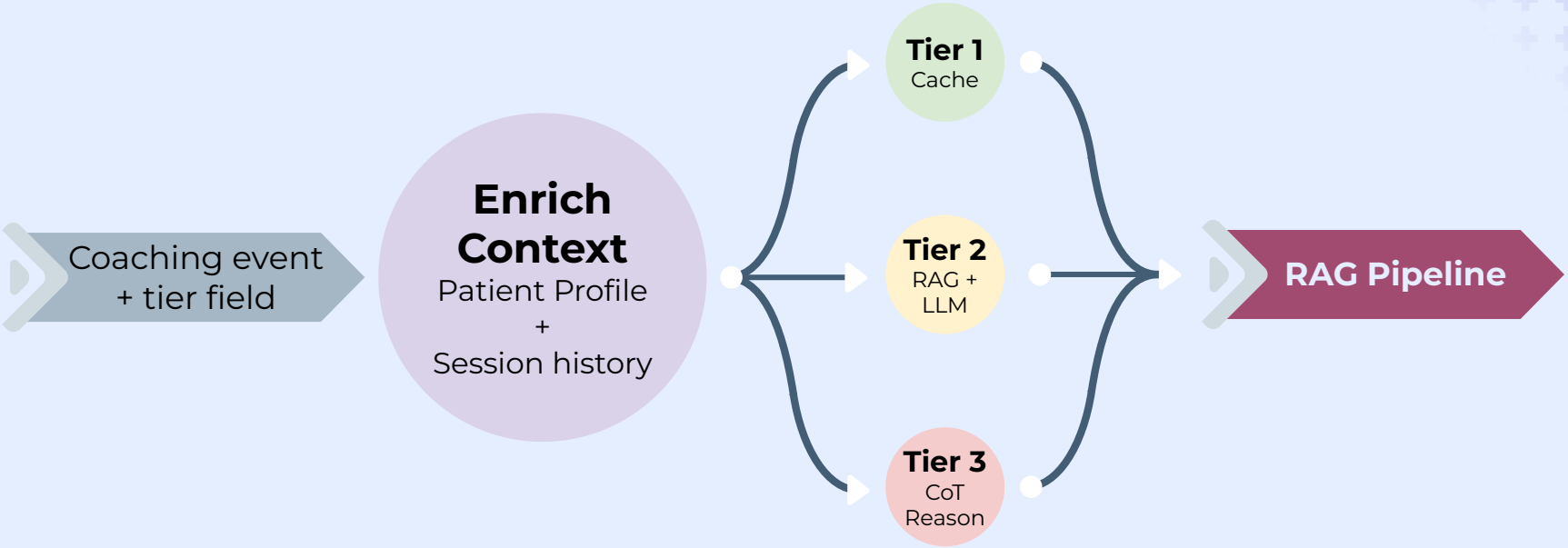
- All 3 models: 100% safety pass rate, professional referral rate, and medical accuracy
- Decision: **Claude Sonnet 4.5** selected for Coaching agent. **Gemma 3:4b** selected for Progress Tracking agent.

Integration Layer: The Intelligence Bridge

- Problem: 15 FPS × 60s = 900 raw predictions/minute — need intelligent filtering
- Four-stage pipeline diagram



Integration Layer: LangGraph Workflow



- Validation results on 1,509 QEVD videos:
  - ~95% noise reduction
  - 27–32% video coverage (persistent error detection)
  - Tier distribution within designed bounds

Multi-Agent System Overview

Computer Vision Module + Integration layer

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Coaching Agent

- Receives structured action features (quality score, mistake type, severity)
- Retrieves clinical guidance via RAG
- Generates real-time exercise feedback + phase report

↓

Phase JSON (structured memory)

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Progress Tracking Agent

- Reads cumulative phase JSONs across sessions
- Analyzes pain trend, quality trend, persistent mistakes
- Generates longitudinal progress report

Coaching Agent — Real-Time Feedback Pipeline

① Session State Machine

- Ingests coaching events every 5s from upstream CV module
- 30s inactivity → exercise ends → exercise-level feedback generated
- 120s inactivity → rehab phase ends → phase report generated + JSON exported

② Two-Level Output

- Exercise feedback: what went wrong, what was good, one correction cue
- Phase report: session overview, top issues, next session recommendations

③ Structured Memory Output

- Exports Phase JSON containing: exercise list, quality scores per event, mistake frequency, pain level, next phase focus
- This JSON is the handoff to Progress Tracking Agent

Progress Tracking Agent: Long-Term Memory & Trend Analysis

① Phase JSON as Inter-Agent Memory

- Coaching Agent writes → Progress Tracker reads
- Each JSON = one rehab phase snapshot
- Multiple JSONs across sessions = longitudinal patient record
- Enables medium-to-long term memory without a database

② What Progress Tracker Analyzes

- Pain trend across phases (numeric + direction)
- Quality score trend per exercise
- Persistent mistakes (appear in 2+ phases → flagged)
- Exercise progression (which exercises added/removed per phase)
- LLM generates a narrative progress report grounded in all phase data

RAG Module — Clinically Grounded Coaching

① Knowledge Base

- Source: physiotherapy textbooks (Kisner & Colby, Houglum) + NHS MSK patient guides
- 1,486 chunks indexed in ChromaDB
- Embedding model: sentence-transformers/all-MiniLM-L6-v2

② Retrieval Strategy

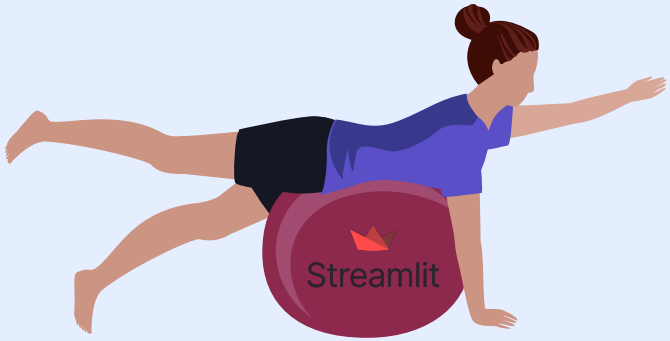
- Query built dynamically from Patient context (condition + rehab phase + patient concern)
- k=3 retrieval — balances relevance and memory footprint on local 4B model
- Clinical context injected into prompt with strict boundary constraints to reduce hallucination

③ Evaluation (No Ground Truth)

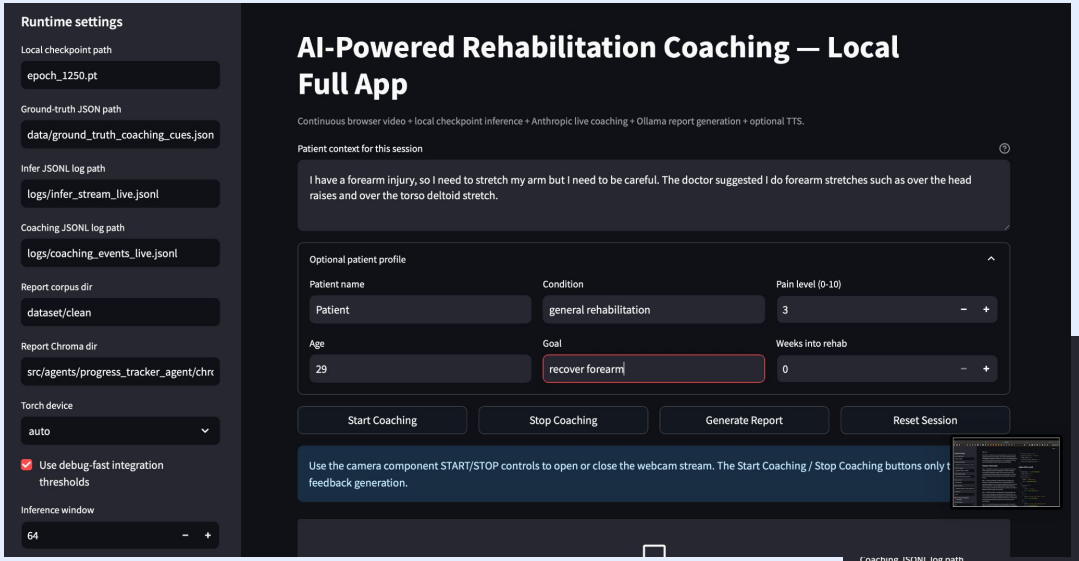
- Precision@k (k=3,5,10): measures whether retrieved chunks are relevant to the condition
- Source Coverage: verifies returned sources cover expected document types
- Response Quality Score: rule-based checklist (exercise recommendation, safety language, appropriate length, motivational tone)

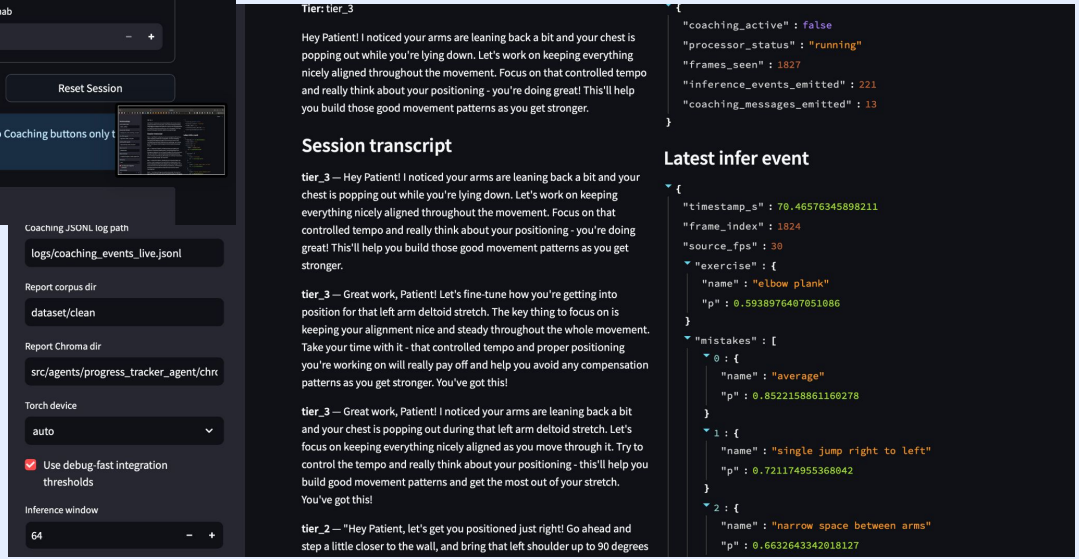
WHOA!

It’s Time for a Streamlit Demo.



Example Demo Output





Example Demo Output

Runtime settings

Local checkpoint path
epoch_1250.pt

Ground-truth JSON path
data/ground_truth_coaching_cues.json

Infer JSONL log path
logs/infer_stream_live.jsonl

Coaching JSONL log path
logs/coaching_events_live.jsonl

Report corpus dir
dataset/clean

Report Chroma dir
src/agents/progress_tracker_agent/chr

Torch device
auto

☑ Use debug-fast integration thresholds

Inference window
64 − +

Deplo

step a little closer to the wall, and bring that left shoulder up to 90 degrees before you start your stretch. You've got this!"

tier_2 — "Nice work! Just remember to keep your chest in a neutral position and lean your whole body forward into that stretch - you don't want to pull your arms back behind you. You've got this!"

tier_2 — "Hey Patient, let's get you positioned just right! Go ahead and step a little closer to the wall, and bring your right arm up to shoulder height. Perfect - now you're all set to start that stretch!"

tier_2 — "Keep your chest neutral and lean your body forward into the stretch, not your arms backward."

tier_2 — "Start standing tall with feet hip-width apart, hands on hips for balance before beginning knee circles."

tier_1 — Maintain proper form

tier_1 — Maintain proper form

tier_2 — "Great job, Patient! Stand nice and tall with both feet planted firmly on the ground, and remember to keep those knees lifted as you do your arm circles. You've got this!"

tier_2 — "Great job, Patient! Just remember to keep both knees lifted off the ground - you've got it! Now let only your head move in that gentle nodding motion. You're doing really well!"

tier_1 — Maintain proper form


```
    "p" : 0.6632643342018127
  }
  3 : {
    "name" : "butt sinking"
    "p" : 0.5560212731361389
  }
]
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  "speed_rps" : 0.85
  "rom_level" : 1
  "height_level" : 3
  "torso_rotation" : 1
  "direction" : "none"
  "no_obvious_issue_p" : 0.00005627355130854994
}
"quality_score" : 0.15883415151329247
"speak_now" : 0
}
```


Session report

Okay, let's get you moving towards recovery! It's great to hear from you, and I understand your cautious approach to your forearm injury and your desire to incorporate stretches like over-the-head raises and deltoid stretches – that's a really smart approach, and we'll build from there.

It sounds like you're experiencing some difficulty getting into those stretches, particularly with your chest popping out and arms leaning back. That's fantastic that you're noticing this – it's a key sign we can adjust. We need to focus on really controlled movement and ensuring you're maintaining good posture throughout.

Deploy

nodding motion. You're doing really well!"

tier_1 — Maintain proper form

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Next Session:

Let's try these exercises, focusing on slow, deliberate movements:

1. **Supported Arm Raise (Right):** Slow raise arm to 45 degrees, hold for 3 seconds, repeat 3 times, 1 repetition. (Focus on controlled movement)
2. **Over-the-Torso Deltoid Stretch (Left):** 3 repetitions, 30-second hold each. (Important: Ensure your chest stays lifted and avoid leaning back.)
3. **Wall Slides:** 3 sets of 8-10 slides – this will gently mobilise the shoulder joint.
4. **Gentle Shoulder Rolls:** 3 repetitions, 10 rotations each direction.

Addressing Your Concern:

You mentioned the doctor suggested these stretches – that's brilliant! We'll tailor them to *your* body and how you're feeling, starting with very gentle ranges of motion.

⚠️ Note: Listen to your body! If you feel any sharp pain, stop immediately and let me know.

You've got this! Recovering your forearm strength and mobility is a realistic goal, and with consistent, careful effort, we'll be seeing improvements in no time. Let's keep building that confidence and getting you back to your full potential. 💪

Pipeline Evaluation and Design Trade-Offs

- Our project focuses on the applied problem thus:
- **Our design** minimizes CV and RAG model size and compute in order to be able to run on budget hardware, as opposed to pushing cutting edge models or techniques
- CV pipeline: We chose TCN over LSTM, GRU and Transformer based alternatives to balance modelling capacity and deployment constraints while still allowing training the model on a single laptop
- RAG pipeline: We chose Claude Sonnet 4.5 over alternatives to maximize generation metrics and especially minimize latency of full RAG generation to still non-ideal <8 seconds

- **Evaluation**: RAG generation refined over 30+ diverse physiotherapy question subsets weighing fact and semantic coverage, safety pass rate and medical accuracy as well as operational metrics
- Dual pipeline testing includes 3 end-to-end physiotherapy scenarios focusing on rehabilitation for:
  a. Rotator cuff injury
  b. ACL recovery
  c. Lower back pain

Individual Contributions

Jason - Mainly worked in the project managing role, and did the CV and RAG research, CV and integration layer design as well as writing the report and iteration assignments

David - Worked on the entire integration layer and agent orchestration. Also contributed towards LLM comprehensive testing and prototyped coaching agent.

Andre - Worked on the entire CV pipeline, organizing the videos, creating the dataset, training CV model, inference back end, front end streamlit UI integrating with all other code

Rongjia - Worked on the multi-agent pipeline, set up the RAG knowledge base for the agents. Set up mid term memory system, bridging the coaching agent and the progress tracker agent.

Limitations & Future Work

- We’ve built a functional end-to-end VPA but it still needs more refinement as well as fine-tuning and testing across different exercise sub-types within our 148 exercise dataset
- Polish UI and session by session patient interaction to optimize how it encourages compliance
- Implement smartphone app version so patients can use it wherever they are esp. on the go
- RAG system design is traditional. Could try graph based RAG to be able to better understand the hierarchy relation between sources as well as textbooks
- With more time, we could fine-tune a local open-source medical LLM ex: Medgemma
    a. No API based multi-second latency, runs locally
    b. Full visibility and ability to modify weights
    c. Newer models like Medgemma are too taxing for even mid-tier laptops

THANKS!

DO YOU HAVE ANY QUESTIONS?

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